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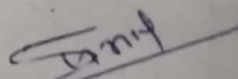
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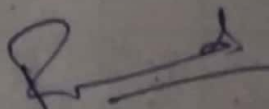
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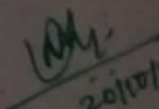
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UNIT-4 (ADVANCE ENGINEERING MATHEMATICS)

INTRODUCTION TO LINEAR PROGRAMMING PROBLEM

LECTURE NO: -26

Introduction to Linear programming Problem:

Linear programming is a simple technique where we depict complex relationships through linear functions and then find the optimum points. So Linear programming is the process of taking various linear inequalities relating to some situation, and finding the "best" value obtainable under those conditions. A typical example would be taking the limitations of materials and labour, and then determining the "best" production levels for maximal profits under those conditions.

In "real life", linear programming is part of mathematics called "optimization techniques". This field of study are used every day in the organization and allocation of resources. Applications of linear programming are everywhere around you. when you have a project delivery you make strategies to make your team work efficiently for on-time delivery.

Let us look at the steps of defining a Linear Programming problem generically:

1. Identify the decision variables
2. Write the objective function
3. Mention the constraints

Explicitly state the non-negativity restriction

For a problem to be a linear programming problem, the decision variables, objective function and constraints all have to be linear functions.

If all the three conditions are satisfied, it is called a **Linear Programming Problem**.

The solution of a linear programming problem reduces to finding the optimum value (largest or smallest, depending on the problem) of the linear expression (called the objective function)

$$f = c_1x_1 + \dots + c_nx_n$$

subject to a set of constraints expressed as inequalities:

$$\begin{aligned} a_{11}x_1 + \dots + a_{1n}x_n &\leq b_1 \\ &\vdots \\ a_{m1}x_1 + \dots + a_{mn}x_n &\leq b_m \text{ with } \forall x_i \geq 0. \end{aligned}$$

The a 's, b 's, and c 's are constants determined by the capacities, needs, costs, profits, and other requirements and restrictions of the problem. The problem can be stated in the matrix form as

$$\text{Optimize } Z = CX,$$

$$\text{Subject to } AX (\leq, \geq, =) b$$

$$X \geq 0$$

Where $C = (c_1, c_2, c_3, \dots, c_n)$, $X = (x_1, x_2, x_3, \dots, x_n)$

$$A = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \dots & a_{mn} \end{pmatrix}, b = \begin{pmatrix} b_1 \\ \dots \\ b_m \end{pmatrix}$$

- (1) Each entry of the column vector b is taken to be non-negative if not, we multiply the constraint by -1 to make it such.
- (2) In LPP, it is a must that the values of all variables x_j be either zero or +ve i.e., all decision variables are non-negative.
- (3) In a general LPP, it is assumed that the number of rows of coefficient matrix A is less than its number of columns.

Basic Definitions and Theorems of LPP

Solution: values of the decision variables $x_i (i=1,2,3,\dots)$ satisfying the constraints of general LPP is known as the solution to that LPP.

Basic Solution: Consider a system $AX=b$ of m equations in n unknown variables ($n > m$) and $r(A)=r(AB)=m$. then none of the equations is redundant.

If any $m \times m$ non-singular matrix whose determinant is non zero is chosen from A and if all the $(n-m)$ variables not associated to with the column of this matrix are set equal to zero the solution to the resulting system of equations is called a basic solution.

In other words, a solution obtained by setting any $(n-m)$ variables to zero is called a solution, provided the determinant of the coefficients to the remaining m variables is not zero. Thus in a basic solution at least $n-m$ variables must vanish.

Basic variables: The m variables associated with the column of the non-singular matrix which may be different from zero are called basic variables.

Non-degenerate basic solution: If none of the basic variable is zero, it is non-degenerate basic solution.

Degenerate basic solution: A one or more basic variables becomes equal to zero, then it is called degenerate basic solution.

Theorem: A necessary and sufficient condition for the existence and non-degeneracy of the basic solutions of $AX=b$ is that every set of m columns of the augmented matrix $[A, b]$ is linearly independent.

Feasible Solution: A feasible solution of a LPP is the set of the variables which satisfies the set of constraints and non-negative restrictions of the problem.

Basic Feasible Solution(BFS): A feasible solution of a general LPP is also a basic solution is called a basic feasible solution.

Non-degenerate BFS: A BFS of a LPP is said to non- degenerate if non of the basic variables is zero.

Degenerate BFS: A BFS of a LPP is said to degenerate if atleast one of the basic variables is zero.

Optimal Feasible Solution: Any BFS which optimizes objective function of a linear programming modal is known as optimal feasible solution of that linear programming modal.

Extreme Points: For any convex set S , a point P in S is an extreme point if each line segment that lies completely in S contains the point P has P as an end point of the line segment.

Extreme points are sometime called corner points because if the set is polygon, the extreme points will be the vertices or corner of the polygon.

Convex set: A set of points of set S , is convex set if the line segments joining any two pairs of points in S is wholly contained in S

Some Important Theorems:

- (1) Intersection of any number of convex sets is a convex set.
- (2) The set of all convex combination of finite number of points $x_1, x_2, \dots, x_n$ is a convex set.
- (3) The set of all feasible solutions of a LPP is a convex set.
- (4) Every BFS of a system $AX=b, X \geq 0$ is an extreme point of a convex set of feasible solution and conversely.
- (5) Atleast one of the extreme points of the convex set of the feasible solution represent the optimal solution.
- (6) If the objective function of a LPP assumes it's optimal value, at more than one extreme points, then every convex combination of these extreme value gives the optimal value of the objective function.

Solutions of a LPP:

In general, we use the following two method to solve a LPP.

1. Graphical Method: This method is generally used when we are dealing two variables only.
2. Simplex Method: This method works all LPP problems. It is a very powerful tool to solve a LPP.

Types of solutions in a LPP: In general a LPP may have,

1. A definite and unique optimal solution
2. An infinite number of solutions
3. An unbounded solution
4. No solution

References:

Advance Engineering Mathematics - H.K.Das, S. Chand

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UNIT-4 (ADVANCE ENGINEERING MATHEMATICS)

SIMPLEX METHOD

LECTURE NO: -27

Simplex Method

Simplex method, also called simplex technique or simplex algorithm in linear programming for solving an optimization problem, typically one involving a function and several constraints expressed as inequalities. The inequalities define a polygonal region (see polygon), and the solution is typically at one of the vertices. The simplex method is a systematic procedure for testing the vertices as possible solutions.

Simplex method also called simplex technique or simplex algorithm was developed by G.B. Dantzig, An American mathematician. Simplex method is suitable for solving linear programming problems with a large number of variables. The method through an iterative process progressively approaches and ultimately reaches to the maximum or minimum values of the objective function.

In geometry, a simplex is a generalization of the notion of a triangle or tetrahedron to arbitrary dimension. Specially, an n -simplex is an n -dimensional polytope with $n+1$ vertices, of which the simplex is the convex hull. For example, a 2-simplex is a triangle, a 3 simplex is a tetrahedron, and a 4-simplex is a pentahedron. A single point may be considered a 0-simplex, and a line segment may be viewed as a 1-simplex. A simplex may be defined as the smallest convex set which contains the given vertices.

Some Definitions:

Slack Variables: If a constraint has a sign $\leq$, then in order to make it an equality we have to add some positive quantity to the left hand side.

The positive variables which are added to left hand side of the constraints to convert them into equalities are

e.g. $x+y+z+s \leq 5$

$x, y, z \geq 0$ can be converted to $x+y+z=5, x,y,z \geq 0$, then s is called the slack variables.

Surplus variables:

If constraints has a sign $\geq$ then in order to make it an equality we have to subtract some positive quantity from the left hand side.

The positive variable which are subtracted from the left hand side of the constraints to convert them into equalities are called the surplus variables.

e.g. $2x+y \geq 200$ can be converted to

$$2x+y-s=200$$

Then s is the surplus variable.

The cost value of slack and surplus variables in the objective function is zero.

Computational Procedure of Simplex Method:

The computational aspect of the simplex procedure is best explained by a simple example.

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Example 1:

Consider the linear programming problem:

$$\text{Maximize } z = 3x_1 + 2x_2$$

$$\text{Subject to } x_1 + x_2 = 4$$

$$x_1 - x_2 = 2$$

$$x_1, x_2 \geq 0$$

Solution:

he steps in simplex algorithm are as follows:

Step 1:

Formulation of the mathematical model:

(i) Formulate the mathematical model of given LPP.

(ii) If objective function is of minimisation type then convert it into one of maximisation by following relationship

$$\text{Minimise } Z = - \text{Maximise } Z^*$$

$$\text{When } Z^* = -Z$$

(iii) Ensure all b_i values [all the right side constants of constraints] are positive. If not, it can be changed into positive value on multiplying both side of the constraints by -1.

In this example, all the b_i (height side constants) are already positive.

(iv) Next convert the inequality constraints to equation by introducing the non-negative slack or surplus variable. The coefficients of slack or surplus variables are zero in the objective function.

In this example, the inequality constraints being '=' only slack variables s_1 and s_2 are needed.

Therefore given problem now becomes:

Therefore given problem now becomes

$$\text{Max } z = 3x_1 + 2x_2 + 0s_1 + 0s_2$$

$$\text{Subject to } x_1 + x_2 + s_1 + 0s_2 = 4$$

$$x_1 - x_2 + 0s_1 + s_2 = 2$$

$$x_1, x_2, s_1, s_2 \geq 0$$

(v) Now present the constraints equations in matrix form :

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

Step 2:

Set up the initial solution.

Write down the coefficients of all the variables in given LPP in the tabular form, as shown in table below to get an initial basic Feasible solution.

$$x_B = B^{-1} b$$

Table 1: Initial Simplex Table

		$C_j \rightarrow$	c_1	c_2	$\dots$	C_n	$0..0-0$
Coefficient of Basic	Variables in Basic	Value of basic Variables	Variables – Variables				
Variables (C_B)	B	$b (=x_B)$	x_1	x_2	$\dots$	x_n	$s_1 s_2 \dots s_n$
C_{B1}	s_1	$x_{B1} = b_1$	a_{11}	a_{12}	a_{1n}	1	0 0
C_{B2}	s_2	$x_{B2} = b_2$	a_{21}	a_{22}	a_{2n}	1	0 0
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
C_{Bm}	s_m	$x_{Bm} = b_m$	a_{m1}	a_{m2}	a_{mn}	0	0 1
$Z = \sum C_{Bi} \cdot X_{Bi}$			$z_j = \sum C_{Bi} \cdot x_j$				
$= (\text{B.V. Coefficients}) \times (\text{Values of B.V.}) \times (j^{\text{th}} \text{ column of data matrix})$			$c_j - z_j$				
			0	0	0	0	0 0
			$c_1 - z_1$	$c_2 - z_2$	$\dots$	$c_n - z_n$	0 0 $\dots$ 0

The first row in table indicates the coefficient c_j of variables in objective function, which remain same in successive tables. These values represent cost or profit per unit of objective function of each of the variables.

The second row gives major column headings for the simple table. Column C_B gives the coefficients of the current basic variables in the objective function. Column x_B gives the current values of the corresponding variables in the basic.

Number a_{ij} represent the rate at which resource $(i-1, 2-m)$ is consumed by each unit of an activity j ($j = 1, 2 \dots n$).

The values z_j represents the amount by which the value of objective function Z would be decreased or increased if one unit of given variable is added to the new solution.

It should be remembered that values of non-basic variables are always zero at each iteration.

So $x_1 = x_2 = 0$ here, column x_B gives the values of basic variables in the first column.

So $s_1 = 4, s_2 = 2$, here; The complete starting feasible solution can be immediately read from table 2 as $s_1 = 4, s_2, x_1 = 0, x_2 = 0$ and the value of the objective function is zero.

$C_j \rightarrow$		3	2	0	0	
Basic Variables	C_B X_B	X_1	X_2	$S_1 (= x_3)$ B_1	$S_2 (=4)$ B_2	MIN Ratio ($\frac{x_B}{x_K}$)
S_1	0 4	1	1	1	0	to be completed in the next step
S_2	0 2	1	-1	0	1	
Z	$Z = C_B X_B$	$\Delta_1 = -3$ $\uparrow$	$\Delta_2 = -2$	$\Delta_3 = 0$	$\Delta_4 = 0$ $\downarrow$	$\Delta_j = z_j - c_j$

Step 3:**Test for optimality:**

Now, proceed to test basic feasible for optimality by the rules given below. This is done by computing the “net evaluation” D_j for variable x_j by the formula

$$\begin{aligned}\Delta_j &= z_j - c_j \\ &= C_B X_j - C_j\end{aligned}\quad \dots(1)$$

Thus

$\begin{aligned}\Delta_1 &= C_B X_1 - C_1 \\ &= (0,0)(1,1) - 3 \\ &= (0 \times 1 + 0 \times 1) - 3 \\ &= -3\end{aligned}$	$\begin{aligned}\Delta_2 &= C_B X_2 - C_2 \\ &= (0,0)(1,-1) - 2 \\ &= (0 \times 1 + 0 \times -1) - 2 \\ &= -2\end{aligned}$	$\begin{aligned}\Delta_3 &= C_B X_3 - C_3 \\ &= (0,0)(1,0) - 0 \\ &= (0 \times 1 + 0 \times 0) - 0 \\ &= 0\end{aligned}$
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Optimality Test:

- (i) If all $\Delta_j = 0$, the solution under test will be optimal.
- (ii) If at least one Δ_j is negative, the solution under test is not optimal, then proceed to improve the solution in step 4.
- (iii) If corresponding to most negative Δ_j , all elements of the column X_j are negative or zero ($= 0$), then the solution under test will be unbounded

Applying this rule for testing the optimality of starting basic feasible solution, it is observed that Δ_1 and Δ_2 both are negative. Hence proceed to improve this solution in step 4.

Step 4:

In order to improve this basic feasible solution, the vector or entering the basis matrix and the vector to be removed from the basis matrix are determined by the following rules, such vectors are usually named as “incoming vector” and “outgoing vector” respectively.

“Incoming Vector”:

The incoming vector X_k is always selected corresponding to the most negative value of Δ_j . (say Δ_k). Here $\Delta_k = \text{Min} (\Delta_1, \Delta_2) = \text{Min} [-3, -2] = -3 = \Delta_1$.

Therefore $k = 1$ and column vector x_1 , must enter the basis matrix. The column x_1 is marked by an upward arrow ($\uparrow$).

“Outgoing Vector”:

The outgoing vector β_r is selected corresponding to the minimum ratio of elements of X_B by the corresponding positive elements of predetermined incoming vector X_k . This rule is called the Minimum Ratio Rule. In mathematical form, this rule is written as,

References:

Advance Engineering Mathematics - H.K. Das, S. Chand

UNIT-4 (ADVANCE ENGINEERING MATHEMATICS)

BIG-M METHOD

LECTURE NO: - 28

Problems on Simplex Method

(1) Minimize $z = x_2 - 3x_3 + 2x_5$

Subject to $3x_2 - 2x_3 + 2x_5 \leq 7$

$$\begin{aligned} -2x_2 + 4x_3 &\leq 12 \\ -4x_2 + 3x_3 + 8x_5 &\leq 10 \\ x_2, x_3, x_5 &\geq 0 \end{aligned}$$

Solution : Equivalently $\max z^1 = -x_2 + 3x_3 - 2x_5$

where $z^1 = -z$, Introducing x_1, x_4 , and x_6 , as slack variable, the constraint equation becomes:

$$\begin{aligned} x_1 + 3x_2 - x_3 + 0x_4 + 2x_5 + 0x_6 &= 7 \\ 0x_1 - 2x_2 + 4x_3 + 0x_4 + 0x_5 + 0x_6 &= 12 \\ 0x_1 - 4x_2 + 3x_3 + 0x_4 + 8x_5 + x_6 &= 10 \end{aligned}$$

Now proceeding as in above example the simplex computation are performed as follows :

		$C_j \rightarrow 0 \quad -1 \quad 3 \quad 0 \quad -2 \quad 0$							
Basic Variables	C_B	X_B	x_1	x_2	x_3	x_4	x_5	x_6	Min Ratio
x_1	0	7	1	3	-1	0	2	0	
x_3	0	12	0	-2	-1 ... 1 ... 0 ... 0				12/4=3 min
x_6	0	10	0	-4	3	0	8	1	10/3
$x_2 = x_3 = x_5 = 0$	$z^1 = C_B X_B = 0$		0	1	-3 ↑	0	2	0	$\leftarrow \Delta_j$
x_1	0	10	1	5/2 ... 0 ... 1/4 ... 2 ... 0 ...					4 ← min
x_3	3	3	0	-1/2	1	1/4	0	0	-
x_6	0	1	0	-5/2	0	-3/4	8	1	-
$x_2 = x_4 = x_5 = 0$	$z^1 = C_B X_B = 9$		0	-1/2* ↑	0	3/4	2	0	$\leftarrow \Delta_j$
x_2	-1	4	2/5	1	0	1/10	4/5	0	
x_3	3	5	1/5	0	1	3/10	2/5	0	
x_6	0	11	1	0	0	-1/2	10	1	
$x_1 = x_4 = x_5 = 0$	$z^1 = C_B X_B = 0$ $z = -11$		1/5	0	0	4/5	12/5	0	$\leftarrow \Delta_j$

Thus optimal solution is $x_2 = 4, x_3 = 5, x_6 = 11$

(2) Maximize $z = 3x_1 + 2x_2$

subject to

$$\begin{array}{rcl} -x_1 & +2x_2 & \leq 4 \\ 3x_1 & +2x_2 & \leq 14 \end{array}$$

$$x_1 - x_2 \leq 3$$

$$x_1, x_2 \geq 0$$

Solution.

First, convert every inequality constraints in the LPP into an equality constraint, so that the problem can be written in a standard form. This can be accomplished by adding a slack variable to each constraint. Slack variables are always added to the less than type constraints.

Converting inequalities to equalities

$$\begin{array}{rclcl} -x_1 & +2x_2 & & +x_3 & =4 \\ 3x_1 & +2x_2 & & +x_4 & =14 \\ x_1 & -x_2 & & +x_5 & =3 \\ x_1, x_2, x_3, x_4, x_5 & \geq 0 & & & \end{array}$$

Where x_3, x_4 and x_5 are slack variables.

Since slack variables represent unused resources, their contribution in the objective function is zero. Including these slack variables in the objective function, we get

$$\text{Maximize } z = 3x_1 + 2x_2 + 0x_3 + 0x_4 + 0x_5$$

Initial basic feasible solution

Now we assume that nothing can be produced. Therefore, the values of the decision variables are zero.

$$x_1 = 0, x_2 = 0, z = 0$$

When we are not producing anything, obviously we are left with unused capacity $x_3 = 4, x_4 = 14, x_5 = 3$

We note that the current solution has three variables (slack variables x_3, x_4 and x_5) with non-zero solution values and two variables (decision variables x_1 and x_2) with zero values. Variables with non-zero values are called basic variables. Variables with zero values are called non-basic variables.

Simplex Method: Table 1

	C_j	3	2	0	0	0	
C_B	Basic variables B	x_1	x_2	x_3	x_4	x_5	Solution values $b (=X_B)$
0	x_3	-1	2	1	0	0	4
0	x_4	3	2	0	1	0	14
0	x_5	1	-1	0	0	1	3
$Z_j - C_j$		-3	-2	0	0	0	

$$\begin{aligned}
 a_{11} &= -1, & a_{12} &= 2, & a_{13} &= 1, & a_{14} &= 0, & a_{15} &= 0, & b_1 &= 4 \\
 a_{21} &= 3, & a_{22} &= 2, & a_{23} &= 0, & a_{24} &= 1, & a_{25} &= 0, & b_2 &= 14 \\
 a_{31} &= 1, & a_{32} &= -1, & a_{33} &= 0, & a_{34} &= 0, & a_{35} &= 1, & b_3 &= 3
 \end{aligned}$$

Calculating values for the index row ($z_j - c_j$)

$$Z_1 - C_1 = (0 \times (-1) + 0 \times 3 + 0 \times 1) - 3 = -3$$

$$Z_2 - C_2 = (0 \times 2 + 0 \times 2 + 0 \times (-1)) - 2 = -2$$

$$Z_3 - C_3 = (0 \times 1 + 0 \times 0 + 0 \times 0) - 0 = 0$$

$$Z_4 - C_4 = (0 \times 0 + 0 \times 1 + 0 \times 0) - 0 = 0$$

$$Z_5 - C_5 = (0 \times 0 + 0 \times 0 + 0 \times 1) - 0 = 0$$

Choose the smallest negative value from $z_j - c_j$ (i.e., -3). So column under x_1 is the key column.

Now find out the minimum positive value

$$\text{Minimum}(14/3, 3/1) = 3.$$

So row x_5 is the key row.

Here, the pivot (key) element = 1 (the value at the point of intersection).

Therefore, x_5 departs and x_1 enters.

We obtain the elements of the next table using the following rules:

1. If the values of $z_j - c_j$ are positive, the inclusion of any basic variable will not increase the value of the objective function. Hence, the present solution maximizes the objective function. If there are more than one negative values, we choose the variable as a basic variable corresponding to which the value of $z_j - c_j$ is least (most negative) as this will maximize the profit.

2. The numbers in the replacing row may be obtained by dividing the key row elements by the pivot element and the numbers in the other two rows may be calculated by using the formula:

New number = old number - (corresponding no. of key row) X (corresponding no. of key column)
pivot element

Calculating values for table 2

x_3 row

$$\begin{aligned} a_{11} &= -1 - 1 \times ((-1)/1) = 0 \\ a_{12} &= 2 - (-1) \times ((-1)/1) = 1 \\ a_{13} &= 1 - 0 \times ((-1)/1) = 1 \\ a_{14} &= 0 - 0 \times ((-1)/1) = 0 \\ a_{15} &= 0 - 1 \times ((-1)/1) = 1 \\ b_1 &= 4 - 3 \times ((-1)/1) = 7 \end{aligned}$$

x_4 row

$$\begin{aligned} a_{21} &= 3 - 1 \times (3/1) = 0 \\ a_{22} &= 2 - (-1) \times (3/1) = 5 \\ a_{23} &= 0 - 0 \times (3/1) = 0 \\ a_{24} &= 1 - 0 \times (3/1) = 1 \\ a_{25} &= 0 - 1 \times (3/1) = -3 \\ b_2 &= 14 - 3 \times (3/1) = 5 \end{aligned}$$

x_1 row

$$\begin{aligned} a_{31} &= 1/1 = 1 \\ a_{32} &= -1/1 = -1 \\ a_{33} &= 0/1 = 0 \\ a_{34} &= 0/1 = 0 \\ a_{35} &= 1/1 = 1 \\ b_3 &= 3/1 = 3 \end{aligned}$$

Table 2

		c_j	3	2	0	0	0	
c_B	Basic variables B	x_1	x_2	x_3	x_4	x_5	Solution values $b (= X_B)$	
0	x_3	0	1	1	0	1	7	
0	x_4	0	5	0	1	-3	5	
3	x_1	1	-1	0	0	1	3	

$$Z_j - C_j \quad 0 \quad -5 \quad 0 \quad 0 \quad 3$$

Calculating values for the index row ($z_j - c_j$)

$$z_1 - c_1 = (0 \times 0 + 0 \times 0 + 3 \times 1) - 3 = 0$$

$$z_2 - c_2 = (0 \times 1 + 0 \times 5 + 3 \times (-1)) - 2 = -5$$

$$z_3 - c_3 = (0 \times 1 + 0 \times 0 + 3 \times 0) - 0 = 0$$

$$z_4 - c_4 = (0 \times 0 + 0 \times 1 + 3 \times 0) - 0 = 0$$

$$z_5 - c_5 = (0 \times 1 + 0 \times (-3) + 3 \times 1) - 0 = 3$$

Key	column	=	x_2	column
Minimum	(7/1,	5/5)	=	1
Key	row	=	x_4	row
Pivot	element	=		5

x_4 departs and x_2 enters.

Calculating values for table 3

x_3 row

$$a_{11} = 0 - 0 \times (1/5) = 0$$

$$a_{12} = 1 - 5 \times (1/5) = 0$$

$$a_{13} = 1 - 0 \times (1/5) = 1$$

$$a_{14} = 0 - 1 \times (1/5) = -1/5$$

$$a_{15} = 1 - (-3) \times (1/5) = 8/5$$

$$b_1 = 7 - 5 \times (1/5) = 6$$

x_2 row

$$a_{21} = 0/5 = 0$$

$$a_{22} = 5/5 = 1$$

$$a_{23} = 0/5 = 0$$

$$a_{24} = 1/5$$

$$a_{25} = -3/5$$

$$b_2 = 5/5 = 1$$

x_1 row

$$a_{31} = 1 - 0 \times (-1/5) = 1$$

$$a_{32} = -1 - 5 \times (-1/5) = 0$$

$$a_{33} = 0 - 0 \times (-1/5) = 0$$

$$a_{34} = 0 - 1 \times (-1/5) = 1/5$$

$$a_{35} = 1 - (-3) \times (-1/5) = 2/5$$

$$b_3 = 3 - 5 \times (-1/5) = 4$$

Don't convert the fractions into decimals, because many fractions cancel out during the process while the conversion into decimals will cause unnecessary complications.

Simplex Method: Final Optimal Table

	c_j	3	2	0	0	0	
c_B	Basic variables B	x_1	x_2	x_3	x_4	x_5	Solution values $b (= X_B)$
0	x_3	0	0	1	$-1/5$	$8/5$	6
2	x_2	0	1	0	$1/5$	$-3/5$	1
3	x_1	1	0	0	$1/5$	$2/5$	4
$z_j - c_j$		0	0	0	1	0	

Since all the values of $z_j - c_j$ are positive, this is the optimal solution.
 $x_1 = 4, x_2 = 1$
 $z = 3 \times 4 + 2 \times 1 = 14.$

References:

Advance Engineering Mathematics - H.K. Das, S. Chand

UNIT-4 (ADVANCE ENGINEERING MATHEMATICS)

TWO PHASE METHOD

LECTURE NO: -29

Two Phase Method

In L.P.P., if some of the constraints have $\geq$ sign, we introduce surplus and artificial variables,

Two-Phase Method Steps (Rule)	
Step-1:	Phase-1 a. Form a new objective function by assigning zero to every original variable (including slack and surplus variables) and -1 to each of the artificial variables. eg. $\text{Max } Z = -A_1 - A_2$ b. Using simplex method, try to eliminate the artificial variables from the basis. c. The solution at the end of Phase-1 is the initial basic feasible solution for Phase-2.
Step-2:	Phase-2 a. The original objective function is used and coefficient of artificial variable is 0 (so artificial variable is removed from the calculation process). b. Then simplex algorithm is used to find optimal solution.

and if $=$ sign we introduce artificial variables to get the starting matrix $B=I_m$.

The artificial variables are fictitious and represent no physical entities. L.P.P. containing artificial variables can be solved mainly two phase method, and Big-M method.

As the name suggests it contains two phases:

Phase 1:

In this phase, we change the objective function. All the cost values, except of -1 each and the objective function is maximized. Constraints are taken as they are given in the problem. The problem is solved by simplex method, once we get rid of any artificial variable from the basis, we don't even write it in the simplex table. When all the artificial variables are removed from the basis and $z_j - c_j \geq 0$ then the value of the new objective function will be zero. We end the phase 1.

Exempl 1.

Minimize $z = -3x_1 + x_2 - 2x_3$

subject to

$$\begin{array}{rclcl}
 x_1 & & +3x_2 & & +x_3 & \leq 5 \\
 2x_1 & & -x_2 & & +x_3 & \geq 2 \\
 4x_1 + 3x_2 - 2x_3 & = & 5 & & &
 \end{array}$$

$$x_1, x_2, x_3 \geq 0$$

Solution.

If the objective function is in minimization form, then convert it into maximization form.

Changing the sense of the optimization

Any linear minimization problem can be viewed as an equivalent linear maximization problem, and vice versa. Specifically:

$$\text{Minimize } \sum_{i=1}^n c_i x_i = \text{Maximize } \sum_{i=1}^n (-c_i) x_i$$

If z is the optimal value of the left-hand expression, then $-z$ is the optimal value of the right-hand expression.

$$\text{Maximize } z = 3x_1 - x_2 + 2x_3$$

subject to

$$\begin{array}{rclcl} x_1 & & +3x_2 & & +x_3 & \leq 5 \\ 2x_1 & & -x_2 & & +x_3 & \geq 2 \\ 4x_1 + 3x_2 - 2x_3 & = & 5 & & & \end{array}$$

$$x_1, x_2, x_3 \geq 0$$

Converting inequalities to equalities

$$\begin{array}{rclcl} x_1 & & +3x_2 & & +x_3 & & +x_4 & = & 5 \\ 2x_1 & & -x_2 & & +x_3 & & -x_5 & = & 2 \\ 4x_1 + 3x_2 - 2x_3 & = & 5 & & & & & & \end{array}$$

$$x_1, x_2, x_3, x_4, x_5 \geq 0$$

Where: x_4 is a slack variable
 x_5 is a surplus variable

The surplus variable x_5 represents the extra units.

Now, if we let x_1 , x_2 and x_3 equal to zero in the initial solution, we will have $x_4 = 5$ and $x_5 = -2$, which is not possible because a surplus variable cannot be negative. Therefore, we need **artificial variables**.

$$\begin{array}{rclcl} x_1 & + & 3x_2 & + & x_3 & + & x_4 & = & 5 \\ 2x_1 & - & x_2 & + & x_3 & - & x_5 & + & A_1 & = & 2 \\ 4x_1 + 3x_2 - 2x_3 + A_2 & = & 5 & & & & & & \end{array}$$

$$x_1, x_2, x_3, x_4, x_5, A_1, A_2 \geq 0$$

Where A_1 and A_2 are artificial variables.

Phase 1 of Two Phase Method

In this phase, we remove the artificial variables and find an initial feasible solution of the original problem. Now the objective function can be expressed as

$$\text{Maximize } 0x_1 + 0x_2 + 0x_3 + 0x_4 + 0x_5 + (-A_1) + (-A_2)$$

subject to

$$\begin{array}{rcl} x_1 + & 3x_2 + & x_3 + & x_4 = & 5 \\ 2x_1 - & x_2 + & x_3 - & x_5 + & A_1 = & 2 \\ 4x_1 + 3x_2 - 2x_3 + A_2 = & 5 \end{array}$$

$$x_1, x_2, x_3, x_4, x_5, A_1, A_2 = 0$$

Initial basic feasible solution

The initial basic feasible solution is obtained by setting $x_1 = x_2 = x_3 = x_5 = 0$

Then we shall have $A_1 = 2$, $A_2 = 5$, $x_4 = 5$

Two Phase Method: Table 1

	c_j	0	0	0	0	0	-1	-1	
c_B	Basic variables B	x_1	x_2	x_3	x_4	x_5	A_1	A_2	θ
0	x_4	1	3	1	1	0	0	0	
-1	A_1	2	-1	1	0	-1	1	0	
-1	A_2	4	3	-2	0	0	0	1	
$Z_j - C_j$		-6	-2	1	0	1	0	0	

Key Minimum (5/1, 2/2, 5/4) = x_1 column
Key row = A_1 row
Pivot element = 2
A1 departs and x_1 enters.

Table 2

Phase 2 of Two Phase Method

The basic feasible solution at the end of **Phase 1** computation is used as the initial basic feasible solution of the problem. The original objective function is introduced in **Phase 2** computation and the usual simplex procedure is used to solve the problem.

Table 3

	c_j	3	-1	2	0	0	
c_B	Basic variables B	x_1	x_2	x_3	x_4	x_5	Solution value $b (= X_B)$
0	x_4	0	0	$33/10$	1	$-9/10$	$33/10$
3	x_1	1	0	$1/10$	0	$-3/10$	$11/10$
-1	x_2	0	1	$-4/5$	0	$2/5$	$1/5$
$Z_j - C_j$		0	0	$-9/10$	0	$-13/10$	

Table 4

	c_j	3	-1	2	0	0	
c_B	Basic variables B	x_1	x_2	x_3	x_4	x_5	Solution value $b (= X_B)$
0	x_4	0	$9/4$	$3/2$	1	0	$15/4$
3	x_1	1	$3/4$	$-1/2$	0	0	$5/4$
0	x_5	0	$5/2$	-2	0	1	$1/2$
$Z_j - C_j$		0	$13/4$	$-7/2$	0	0	

Two Phase Method: Final Optimal Table

	c_j	3	-1	2	0	0	
c_B	Basic variables B	x_1	x_2	x_3	x_4	x_5	Solution value $b (= X_B)$
2	x_3	0	$3/2$	1	$2/3$	0	$5/2$
3	x_1	1	$3/2$	0	$1/3$	0	$5/2$
0	x_5	0	$11/2$	0	$4/3$	1	$11/2$
$Z_j - C_j$		0	$17/2$	0	$7/3$	0	

An optimal policy is $x_1 = 5/2$, $x_2 = 0$, $x_3 = 5/2$. The associated optimal value of the objective function is $z = 3 \times (5/2) - 0 + 2 \times (5/2) = 25/2$.

Example 2.

Find solution using Two-Phase method

$x_1=2113$, $x_2=1013$, Min $Z=3113$

References:

Advance Engineering Mathematics - H.K. Das, S. Chand

UNIT-4 (ADVANCE ENGINEERING MATHEMATICS)

DUALITY IN LPP

LECTURE NO: -30

Duality in LPP:

Every LPP called the **primal** is associated with another LPP called **dual**. Either of the problems is primal with the other one as dual. The optimal solution of either problem reveals the information about the optimal solution of the other.

Let the primal problem be

$$\text{Max } Z_x = c_1x_1 + c_2x_2 + \dots + c_nx_n$$

Subject to restrictions

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \leq b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n \leq b_2$$

.

.

.

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n \leq b_n$$

and

$$x_1 \geq 0, x_2 \geq 0, \dots, x_n \geq 0$$

The corresponding dual is defined as

$$\text{Min } Z_w = b_1w_1 + b_2w_2 + \dots + b_mw_m$$

Subject to restrictions

$$a_{11}w_1 + a_{21}w_2 + \dots + a_{m1}w_m \geq c_1$$

$$a_{12}w_1 + a_{22}w_2 + \dots + a_{m2}w_m \geq c_2$$

.

.

.

$$a_{1n}w_1 + a_{2n}w_2 + \dots + a_{mn}w_m \geq c_n$$

and

$$w_1, w_2, \dots, w_m \geq 0$$

Matrix Notation of Primal

$$\text{Max } Z_x = CX$$

Subject to

$$AX \leq b \text{ and } X \geq 0$$

Dual

$$\text{Min } Z_w = b^T W$$

Subject to

$$A^T W \geq C^T \text{ and } W \geq 0$$

Notes By: -Sunil kumar Sharma (Assistant Professor-Maths Department)

Important characteristics of Duality

1. Dual of dual is primal
2. If either the primal or dual problem has a solution then the other also has a solution and their optimum values are equal.
3. If any of the two problems has an infeasible solution, then the value of the objective function of the other is unbounded.
4. The value of the objective function for any feasible solution of the primal is less than the value of the objective function for any feasible solution of the dual.
5. If either the primal or dual has an unbounded solution, then the solution to the other problem is infeasible.
6. If the primal has a feasible solution, but the dual does not have then the primal will not have a finite optimum solution and vice versa.

Advantages and Applications of Duality

7. Sometimes dual problem solution may be easier than primal solution, particularly when the number of decision variables is considerably less than slack / surplus variables.
8. In the areas like economics, it is highly helpful in obtaining future decision in the activities being programmed.
9. In physics, it is used in parallel circuit and series circuit theory.
10. In game theory, dual is employed by column player who wishes to minimize his maximum loss while his opponent i.e. Row player applies primal to maximize his minimum gains. However, if one problem is solved, the solution for other also can be obtained from the simplex tableau.
11. When a problem does not yield any solution in primal, it can be verified with dual.
12. Economic interpretations can be made and shadow prices can be determined enabling the managers to take further decisions.

Steps for a Standard Primal Form

Step 1 – Change the objective function to Maximization form

Step 2 – If the constraints have an inequality sign ' $\geq$ ' then multiply both sides by -1 and convert the inequality sign to ' $\leq$ '.

Step 3 – If the constraint has an '=' sign then replace it by two constraints involving the inequalities going in opposite directions. For example $x_1 + 2x_2 = 4$ is written as

$$x_1 + 2x_2 \leq 4$$

$$x_1 + 2x_2 \geq 4 \text{ (using step2)} \rightarrow -x_1 - 2x_2 \leq -4$$

Step 4 – Every unrestricted variable is replaced by the difference of two non-negative variables.

Step 5 – We get the standard primal form of the given LPP in which.

Notes By: -Sunil kumar Sharma (Assistant Professor-Maths Department)

- Alc1. Constraints have ' $\leq$ ' sign, where the objective function is of maximization form.

1. All constraints have ' $\geq$ ' sign, where the

Rules for Converting any Primal into its Dual

1. Transpose the rows and columns of the constraint co-efficient.
2. Transpose the co-efficient ($c_1, c_2, \dots, c_n$) of the objective function and the right side constants ($b_1, b_2, \dots, b_n$)
3. Change the inequalities from ' $\leq$ ' to ' $\geq$ ' sign.
4. Minimize the objective function instead of maximizing it.

Example Problems

Example 1. Write the dual of the given problems

$$\text{Min } Z_x = 2x_2 + 5x_3$$

Subject to

$$x_1 + x_2 \geq 2$$

$$2x_1 + x_2 + 6x_3 \leq 6$$

$$x_1 - x_2 + 3x_3 = 4$$

$$x_1, x_2, x_3 \geq 0$$

Solution

Primal

$$\text{Max } Z'_x = -2x_2 - 5x_3$$

Subject to

$$-x_1 - x_2 \leq -2$$

$$2x_1 + x_2 + 6x_3 \leq 6$$

$$x_1 - x_2 + 3x_3 \leq 4$$

$$-x_1 + x_2 - 3x_3 \leq -4$$

$$x_1, x_2, x_3 \geq 0$$

Dual

Min $Z_w = -2w_1 + 6w_2 + 4w_3 - 4w_4$ Subject to

$$\begin{aligned} -w_1 + 2w_2 + w_3 - w_4 &\geq 0 \\ -w_1 + w_2 - w_3 + w_4 &\geq -2 \\ 6w_2 + 3w_3 - 3w_4 &\geq -5 \\ w_1, w_2, w_3, w_4 &\geq 0 \end{aligned}$$

Example 2

Min $Z_x = 3x_1 - 2x_2 + 4x_3$

Subject to

$$\begin{aligned} 3x_1 + 5x_2 + 4x_3 &\geq 7 \\ 6x_1 + x_2 + 3x_3 &\geq 4 \\ 7x_1 - 2x_2 - x_3 &\geq 10 \\ x_1 - 2x_2 + 5x_3 &\geq 3 \\ 4x_1 + 7x_2 - 2x_3 &\geq 2 \\ x_1, x_2, x_3 &\geq 0 \end{aligned}$$

Solution

Primal

Max $Z'_x = -3x_1 + 2x_2 - 4x_3$

Subject to

$$\begin{aligned} -3x_1 - 5x_2 - 4x_3 &\leq -7 \\ -6x_1 - x_2 - 3x_3 &\leq -4 \\ -7x_1 + 2x_2 + x_3 &\leq -10 \\ -x_1 + 2x_2 - 5x_3 &\leq -3 \\ -4x_1 - 7x_2 + 2x_3 &\leq -2 \\ x_1, x_2, x_3 &\geq 0 \end{aligned}$$

Dual

Min $Z_w = -7w_1 - 4w_2 - 10w_3 - 3w_4 - 2w_5$

Subject to

$$\begin{aligned} -3w_1 - 6w_2 - 7w_3 - w_4 - 4w_5 &\geq -3 \\ -5w_1 - w_2 + 2w_3 + 2w_4 - 7w_5 &\geq 2 \\ -4w_1 - 3w_2 + w_3 - 5w_4 + 2w_5 &\geq -4 \\ w_1, w_2, w_3, w_4, w_5 &\geq 0 \end{aligned}$$

Example 3

Max $Z = 2x_1 + 3x_2 + x_3$

Subject to

$$\begin{aligned} 4x_1 + 3x_2 + x_3 &= 6 \\ x_1 + 2x_2 + 5x_3 &= 4 \\ x_1, x_2 &\geq 0 \end{aligned}$$

Solution

Notes By: -Sunil kumar Sharma (Assistant Professor-Maths Department)

Primal

$$\text{Max } Z_x = 2x_1 + 3x_2 + x_3$$

Subject to

$$4x_1 + 3x_2 + x_3 \leq 6$$

$$-4x_1 - 3x_2 - x_3 \leq -6$$

$$x_1 + 2x_2 + 5x_3 \leq 4$$

$$-x_1 - 2x_2 - 5x_3 \leq -4$$

$$x_1 \geq 0, x_2 \geq 0.$$

Dual

$$\text{Min } Z_w = 6w_1 - 6w_2 + 4w_3 - w_4$$

Subject to

$$4w_1 - 4w_2 + w_3 - w_4 \geq 2$$

$$3w_1 - 3w_2 + 2w_3 - 2w_4 \geq 3$$

$$w_1 - w_2 + 5w_3 - 5w_4 \geq 1$$

$$w_1, w_2, w_3, w_4 \geq 0$$

Example 4

$$\text{Min } Z_x = x_1 + x_2 + x_3$$

Subject to

$$x_1 - 3x_2 + 4x_3 = 5$$

$$x_1 - 2x_2 \leq 3$$

$$2x_2 - x_3 \geq 4$$

$$x_1, x_2 \geq 0, x_3 \text{ is unrestricted in sign}$$

Solution

Primal

$$\text{Max } Z' = -x_1 - x_2 - x_3' + x_3''$$

Subject to

$$x_1 - 3x_2 + 4(x_3' - x_3'') \leq 5$$

$$-x_1 + 3x_2 - 4(x_3' - x_3'') \leq -5$$

$$x_1 - 2x_2 \leq 3$$

$$-2x_2 + x_3' - x_3'' \leq -4$$

$$x_1, x_2, x_3', x_3'' \geq 0$$

UNIT-4 (ADVANCE ENGINEERING MATHEMATICS)

APPLICATION TO LINEAR PROGRAMMING TRANSPORTATION METHOD

LECTURE NO: -31

Application to Linear Programming to Transportation Problem

The objective of transportation problem is to determine the amount to be transported from each origin to each destinations such that the total transportation cost is minimized.

Transportation Method of Linear programming:

The Transportation problem is basically a L.P. problem that can be solved by the regular simplex method. However, the special structure of the problem, enable us to use the solution procedure, called the transportation technique, which is more efficient. In general, the transportation problem deals with the determination of a minimum cost plan for transporting a single commodity from a number of sources (e.g. warehouses) to a number of destinations (e.g. shopping malls). It can cover practical situations in the area of inventory control, employment-scheduling etc.

The Structure of the Problem

Let there be m origins and n destinations. Let the amount of supply at the i th origin is a_i . Let the demand at j th destination is b_j .

The cost of transporting one unit of an item from origin i to destination j is c_{ij} and is known for all combinations (i,j) . Quantity transported from origin i to destination j be x_{ij}

Transportation modal, infact can be viewed as a network with m sources and n destinations. The line joining source and a destination represents the route through which the commodity is transported. The amount of supply are at source i is a_i and demand in destination j is b_j . The unit transportation cost between source i and destination j is c_{ij} . Let x_{ij} represent the amount transported from source i to destination j , then Lp problem representing the transportation The objective is to determine the quantity x_{ij} to be transported over all routes (i,j) so as to minimize the total transportation cost. The supply limits at the origins and the demand requirements at the destinations must be satisfied.

The above transportation problem can be written in the following tabular form:

		Destinations					supply
		1	2	3	...	n	
Origins	1	(x_{11}) C_{11}	(x_{12}) C_{12}	(x_{13}) C_{13}	...	(x_{1n}) C_{1n}	a_1
	2	(x_{21}) C_{21}	(x_{22}) C_{22}	(x_{23}) C_{23}	...	(x_{2n}) C_{2n}	a_2
	⋮		⋮	⋮	⋮	⋮	⋮
	m	(x_{m1}) C_{m1}	(x_{m2}) C_{m2}	(x_{m3}) C_{m3}	...	(x_{mn}) C_{mn}	a_m
demand		b_1	b_2	b_3	...	b_n	

Table-10.1

Now the linear programming model representing the transportation problem is given by

The objective function is Minimize $Z = \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij}$ Subject to the constraints

$$\sum_{j=1}^n x_{ij} = a_i, i=1,2,\dots,m \text{ (supply constraints)}$$

$$\sum_{i=1}^m x_{ij} = b_j, j=1,2,\dots,n \text{ (demand constraints)}$$

$$x_{ij} \geq 0 \text{ for all } i,j. \text{ (non-negative restrictions)}$$

Some Definitions

Feasible Solution: A feasible solution to a transportation problem is a set of non-negative values $x_{ij}(i=1,2,\dots,m, j=1,2,\dots,n)$ that satisfies the constraints.

Basic Feasible Solution: A feasible solution is called a basic feasible solution if it contains not more than $m+n-1$ allocations, where m is the number of rows and n is the number of columns in a transportation problem.

Optimal Solution: Optimal Solution is a feasible solution (not necessarily basic) which optimizes (minimize) the total transportation cost.

Non degenerate basic feasible Solution: If a basic feasible solution to a transportation problem contains exactly $m+n-1$ allocations in independent positions, it is called a Non degenerate basic feasible solution. Here m is the number of rows and n is the number of columns in a transportation problem.

Degeneracy : If a basic feasible solution to a transportation problem contains less than $m+n-1$ allocations, it is called a degenerate basic feasible solution. Here m is the number of rows and n is the number of columns in a transportation problem.

Obtaining the initial feasible solution, which means identifying the solution that satisfies the requirements of demand and supply. There are several methods through which the initial feasible solution can be obtained; these are:

- North-West Corner
- Least Cost Method
- Vogel's Approximation Method

Note: It is to be ensured that the number of cells occupied should be equal to $m+n-1$, where " m " is the number of rows while " n " is the number of columns.

2. **Testing the optimality of the initial feasible solution.** Once the feasible solution is obtained, the next step is to check whether it is optimum or not. There are two methods used for testing the optimality:
 - Stepping-stone Method
 - Modified Distribution Method (MODI)
3. The final step is to revise the solution until the optimum solution is obtained.

The two most common objectives of transportation problem could be: i) maximize the profit of transporting " n " units of product to the destination " y ", ii) Minimize the cost of shipping " n " units of product to the destination " y ". transportation cost is minimum.

Least Cost Method

Definition: The **Least Cost Method** is another method used to obtain the initial feasible solution for the transportation problem. Here, the allocation begins with the cell which has the minimum cost. The lower cost cells are chosen over the higher-cost cell with the objective to have the least cost of transportation.

The Least Cost Method is considered to produce more optimal results than the North-west Corner because it considers the shipping cost while making the allocation, whereas the North-West corner method only considers the availability and supply requirement and allocation begin with the extreme left corner, irrespective of the shipping cost.

Let's understand the concept of Least Cost method through a problem given below:

Source \ To	D	E	F	Supply
A	5	8	4	50
B	6	6	3	40
C	3	9	6	60
Demand	20	95	35	150

In the given matrix, the supply of each source A, B, C is given Viz. 50units, 40 units, and 60 units respectively. The weekly demand for three retailers D, E, F i.e. 20 units, 95 units and 35 units is given respectively. The shipping cost is given for all the routes.

The minimum transportation cost can be obtained by following the steps given below:

Source \ To	D	E	F	Supply
A	5	8 50	4	50
B	6	6 5	3 35	40
C	3 20	9 40	6	60
Demand	20	95	35	150

North-West Corner Rule

Definition: The **North-West Corner Rule** is a method adopted to compute the initial feasible solution of the transportation problem. The name North-west corner is given to this method because the basic variables are selected from the extreme left corner.

The concept of North-West Corner can be well understood through a transportation problem given below:

Source \ To	D	E	F	Supply
A	5	8	4	50
B	6	6	3	40
C	3	9	6	60
Demand	20	95	35	150

In the table, three sources A, B and C with the production capacity of 50 units, 40 units, 60 units of product respectively is given. Every day the demand of three retailers D, E, F is to be furnished with at least 20 units, 95 units and 35 units of product respectively. The transportation costs are also given in the matrix.

The prerequisite condition for solving the transportation problem is that demand should be equal to the supply. In case the demand is more than supply, then dummy origin is added to the table. The supply of dummy origin will be equal to the difference between the total supply and total demand. The cost associated with the dummy origin will be zero.

Similarly, in case the supply is more than the demand, then dummy source is created whose demand will be equivalent to the difference between supply and demand. Again the cost associated with the dummy source will be zero.

Once the demand and supply are equal, the following procedure is followed:

Source \ To	D	E	F	Supply
A	5 (20)	8 (30)	4	50
B	6	6 (40)	3	40
C	3	9 (25)	6 (35)	60
Demand	20	95	35	150

Select the north-west or extreme left corner of the matrix, assign as many units as possible to cell AD, within the supply and demand constraints. Such as 20 units are assigned to the first cell, that satisfies the demand of destination D while the supply is in surplus.

Now move horizontally and assign 30 units to the cell AE. Since 30 units are available with the source A, the supply gets fully saturated.

Now move vertically in the matrix and assign 40 units to Cell BE. The supply of source B also gets fully saturated.

Again move vertically, and assign 25 units to cell CE, the demand of destination E is fulfilled.

Move horizontally in the matrix and assign 35 units to cell CF, both the demand and supply of origin and destination gets saturated. Now the total cost can be computed.

The Total cost can be computed by multiplying the units assigned to each cell with the concerned transportation cost. Therefore,

$$\text{Total Cost} = 20 \times 5 + 30 \times 8 + 40 \times 6 + 25 \times 9 + 35 \times 6 = \text{Rs } 1015$$

Example: Determine an initial basic feasible solution to the following transportation problem.

	$D_1 D_2 D_3 D_4$				
C_1	6	4	1	5	14
C_2	8	9	2	7	16
C_3	4	3	6	2	5

Solution

Given

	D_1	D_2	D_3	D_4	
C_1	6	4	1	5	14
C_2	8	9	2	7	16
C_3	4	3	6	2	5

6 10 15 4

First allocation

 $D_1 D_2 D_3 D_4$

C_1	(6) 6	4	1	5	14/8
C_2	8	9	2	7	16
C_3	4	3	6	2	5
	6/0	10	15	4	

Second allocation

(6) 6	(8) 4	1	5	14/8/0
8	9	2	7	16
4	3	6	2	5
6/0	10/2	15	4	

Third allocation

(6) 6	(8) 4	1	5	14/8/0
8	(2) 9	2	7	16/14
4	3	6	2	5
6/0	10/2/0	15	4	

Fourth allocation

(6) 6	(8) 4	1	5	14/8/0
8	(2) 9	(14) 2	7	16/14/0
4	3	6	2	5
6/0	10/2/0	15/1	4	

Fifth allocation

(6) 6	(8) 4	1	5	14/8/0
8	(2) 9	(14) 2	7	16/14/0
4	3	(1) 6	2	5/4
6/0	10/2/0	15/1/0	4	

Final allocation

(6) 6	(8) 4	1	5	14/8/0
8	(2) 9	(14) 2	7	16/14/0
4	3	(1) 6	(4) 2	5/4/0
6/0	10/2/0	15/1/0	4/0	

The Associated objective function value is

$$(6 \times 6) + (8 \times 4) + (2 \times 9) + (14 \times 2) + (1 \times 6) + (4 \times 2)$$

$$= 36 + 32 + 18 + 28 + 6 + 8$$

$$= \$128$$

B. Least Cost Method

6	4	1	5	14
8	9	2	7	16
4	3	6	2	5
6	10	15	4	

Solution:

Given transportation problem is

6	4	1	5	14
8	9	2	7	16
4	3	6	2	5
6	10	15	4	

First allocation

6	4	(14) 1	5
8	9	2	7
4	3	6	2
6	10	15/1	4

14/0

16

5

Second allocation

6	4	(14) 1	5	14/0
8	9	(1) 2	7	16/15
4	3	6	2	5
6	10	15/1/0	4	

Third allocation

6	4	(14) 1	5	14/0
8	9	(1) 2	7	16/15
4	3	6	(4) 2	5/1
6	10	15/1/0	4/0	

Fourth allocation

6	4	(14) 1	5	14/0
8	9	(1) 2	7	16/15
4	(1) 3	6	(4) 2	5/1/0
610/9	15/1/0	4/0		

References:

Advance Engineering Mathematics - H.K. Das, S. Chand

Notes By: -Sunil kumar Sharma (Assistant Professor-Maths Department)

UNIT-4 (ADVANCE ENGINEERING MATHEMATICS)

PROBLEMS ON TRANSPORTATION

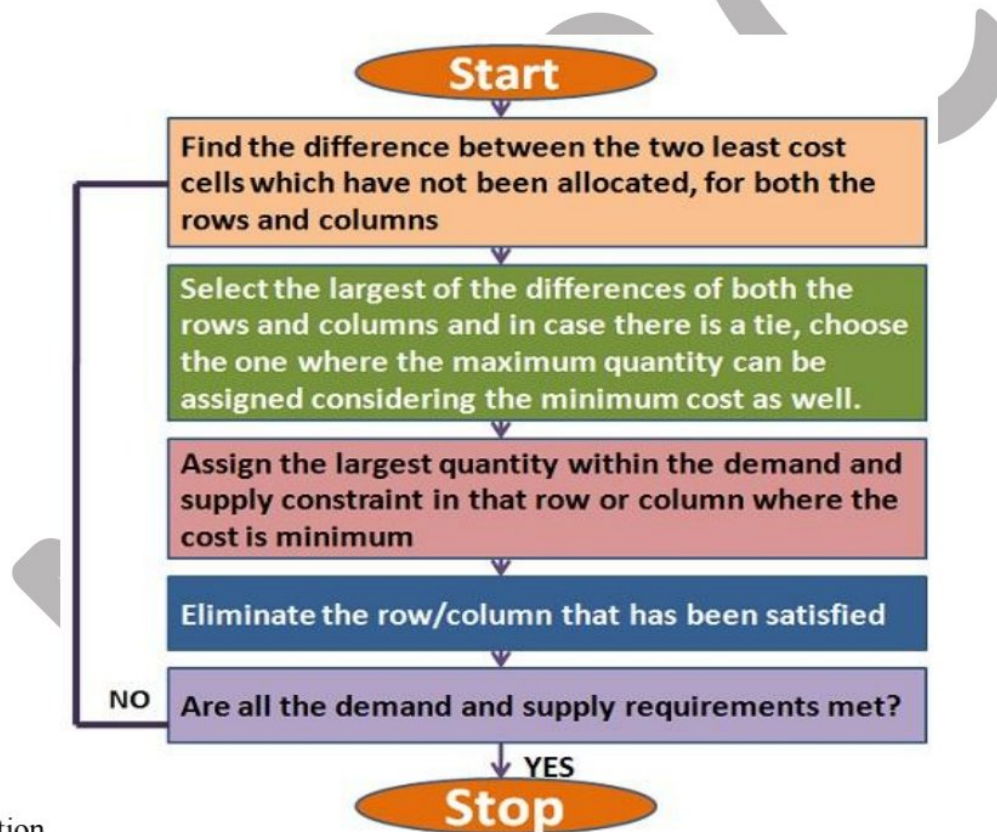
LECTURE NO: -32

Problems on Transportation

Vogel's Approximation Method

Definition: The **Vogel's Approximation Method** or **VAM** is an iterative procedure calculated to find out the initial feasible solution of the transportation problem. Like Least cost Method, here also the shipping cost is taken into consideration, but in a relative sense.

This method gives a very good initial feasible solution, which sometimes is the optimal solution. Or it gives us an initial feasible solution which in a very few iterations gives us the optimal



solution.

The following is the flow chart showing the steps involved in solving the transportation problem using the Vogel's Approximation

Method:

The concept of Vogel's Approximation Method can be well understood through an illustration given below:

First of all the difference between two least cost cells are calculated for each row and column, which can be seen in the iteration given for each row and column. Then the largest difference is selected, which is 4 in this case. So, allocate 20 units to cell BD, since the minimum cost is to be chosen for the allocation. Now, only 20 units are left with the source B.

From \ To	D	E	F	Supply	Iteration-I
A	6	4	1	50	3
B	3 (20)	8	7	40	(4)
C	4	4	2	60	2
Demand	20	95	35	150	
Iteration-I	1	0	1		

Column D is deleted, again the difference between the least cost cells is calculated for each row and column, as seen in the iteration below. The largest difference value comes to be 3, so allocate 35 units to cell AF and 15 units to the cell AE. With this, the Supply and demand of source A and origin F gets saturated, so delete both the row A and Column F.

From \ To	E	F	Supply	Iteration-II
A	4 (15)	1 (35)	50	(3)
B	8	7	20	1
C	4	2	60	2
Demand	95	35	150	
Iteration-II	0	1		

1. Now, single column E is left, since no difference can be found out, so allocate 60 units to the cell CE and 20 units to cell BE, as only 20 units are left with source B. Hence the demand and supply

From \ To	E	Supply
B	8 20	20
C	4 60	60
Demand	80	150

are completely met.

Now the total cost can be computed, by multiplying the units assigned to each cell with the cost concerned. Therefore,

$$\text{Total Cost} = 20 \times 3 + 35 \times 1 + 15 \times 4 + 60 \times 4 + 20 \times 8 = \text{Rs } 555$$

Note: Vogel's Approximation Method is also called as **Penalty Method** because the difference costs chosen are nothing but the penalties of not choosing the least cost routes.

Modified Distribution Method

Initial feasible solution so obtained is tested for optimality. It is important to note that the optimality test is applicable to a F.S. consisting of $m+n+1$ allocation in independent positions.

To test the optimality, we have to consider all the empty cells (where no allocation is made). Take any one of these cells and allocate +1 unit to this cell and make the necessary adjustments to maintain the row and column sum. Calculate the net change in total cost resulting from this adjustment (this is called the evaluation for the empty cell). If the cell evaluation for every empty cell is positive the adjustment will increase the total cost. Thus, if every cell evaluation (for empty cell) is either positive or zero, we can not decrease the total cost and the solution under test is the required optimal solution.

Definition: The **Modified Distribution Method** or **MODI** is an efficient method of checking the optimality of the initial feasible solution.

The concept of MODI can be further comprehended through an illustration given below:

1. Initial basic feasible solution is given

TO From	D	E	F	Supply
A	6 (20)	4 (30)	1	50
B	3	8 (40)	7	40
C	4	4 (25)	2 (35)	60
Demand	20	95	35	150

Optimality: $m+n-1 = 3+3-1 = 5$

below: **Total Cost:** $20 \times 6 + 30 \times 4 + 40 \times 8 + 25 \times 4 + 35 \times 2 = 730$

2. Now, calculate the values of u_i and v_j by using the equation: $u_i + v_j = C_{ij}$

Substituting the value of u_1 as 0

$$u_1 + v_1 = C_{11}, \quad 0 + v_1 = 6 \quad \text{or} \quad v_1 = 6$$

$$u_1 + v_2 = C_{12}, \quad 0 + v_2 = 4 \quad \text{or} \quad v_2 = 4$$

$$u_2 + v_2 = C_{22}, \quad u_2 + 4 = 8 \quad \text{or} \quad u_2 = 4$$

$$u_3 + v_2 = C_{32}, \quad u_3 + 4 = 4 \quad \text{or} \quad u_3 = 0$$

$$u_3 + v_3 = C_{33}, \quad 0 + v_3 = 2 \quad \text{or} \quad v_3 = 2$$

	D	E	F	Supply	u_i
A	6 (20)	4 (30)	1	50	0
B	3	8 (40)	7	40	4
C	4	4 (25)	2 (35)	60	0
Demand	20	95	35	150	
v_j	6	4	2		

=2

3. Next step is to calculate the opportunity cost of the unoccupied cells (AF, BD, BF, CD) by using the following formula: $C_{ij} - (u_i + v_j)$

AF = $C_{13} - (u_1 + v_3) = 1 - (0 + 2) = -1$ or 1
 BD = $C_{21} - (u_2 + v_1) = 3 - (4 + 6) = -7$ or 7
 BF = $C_{23} - (u_2 + v_3) = 7 - (4 + 2) = 1$ or -1
 CD = $C_{31} - (u_3 + v_1) = 4 - (0 + 6) = -2$ or 2

4. Choose the largest positive opportunity cost, which is 7 and draw a closed path, as shown in the matrix below. Start from the unoccupied cell and assign "+" or "-" sign alternatively. Therefore, The most favored cell is BD, assign as many units as

	V1	V2	V3	V4		
	D	E	F	Supply	ui	
A	20 -6	30 +4	1	50	0	u1
B	+3	40 -8	7	40	4	u2
C	4	25 4	35 2	60	0	u3
Demand	20	95	35	150		
vi	6	4	2			

possible.

5. The matrix below shows the maximum allocation to the cell BD, and that number of units are added to the cell with a positive sign and subtracted from the cell with a negative

	V1	V2	V3	V4		
	D	E	F	Supply	ui	
A	6	40 4	1	50	0	u1
B	20 3	20 8	7	40	4	u2
C	4	25 4	35 2	60	0	u3
Demand	20	95	35	150		
vi	6	4	2			

sign.

6. Again, repeat the steps from 1 to 4 i.e. find out the opportunity costs for each unoccupied cell and assign the maximum possible units to the cell having the largest opportunity cost. This process will go on until the optimum solution is reached.

The Modified distribution method is an improvement over the stepping stone method since; it can be applied more efficiently when a large number of sources and destinations are involved, which becomes quite difficult or tedious in case of stepping stone method.

Modified distribution method reduces the number of steps involved in the evaluation of empty cells, thereby minimizes the complexity and gives a straightforward computational scheme through which the opportunity cost of each empty cell can be determined.

References:

Advance Engineering Mathematics - H.K. Das, S. Chand

UNIT-4 (ADVANCE ENGINEERING MATHEMATICS)

ASSIGNMENT PROBLEM

LECTURE NO: -33

Assignment problem:

It is a special type of LPP, in which we assign n jobs to n persons, each person being exactly one job to get the best result. Here we make the gth assumption that each each person can perform each job but with the varying degree of efficiency.

In assignment problem we assign m jobs to n persons, each person being assigned exactly one job to get the best results. Here we make the gth assumption that each person can perform job but with the varying degree of efficiencies.

Example, the munciple committee of a city has a fleet of n tractors located at different places in the city. It is desired to trailers lying at different places in the city. The problem is to assign each of the n tractors to the corresponding trailers in such a way that a given measure of effectiveness is optimized.



1. Assignment Model:

Suppose there are n facilitates and n jobs it is clear that in this case, there will be n assignments. Each facility or say worker can perform each job, one at a time. But there should be certain procedure by which assignment should be made so that the profit is maximized or the cost or time is minimized.

Table 1

Worker	Job of Work					
	1	2	3	4	j th	n
1.	Co_{11}	Co_{12}	Co_{13}	Co_{14}		Co_m
2.	Co_{21}	Co_{22}	Co_{23}	Co_{24}		Co_{2n}
3.	Co_{31}	Co_{32}	Co_{33}	Co_{34}		Co_{3n}
j^{th}				Co_{ij}		
n^{th}	Co_{n1}	Co_{n2}	Co_{n3}	Co_{n4}		Co_{nm}

In the table, Co_{ij} is defined as the cost when j^{th} job is assigned to i^{th} worker. It may be noted here that this is a special case of transportation problem when the number of rows is equal to number of columns.

Mathematical Formulation:

Any basic feasible solution of an Assignment problem consists $(2n - 1)$ variables of which the $(n - 1)$ variables are zero, n is number of jobs or number of facilities. Due to this high degeneracy, if we solve the problem by usual transportation method, it will be a complex and time consuming work. Thus a separate technique is derived for it. Before going to the absolute method it is very important to formulate the problem.

Suppose x_{ij} is a variable which is defined as

1 if the i^{th} job is assigned to j^{th} machine or facility

0 if the i^{th} job is not assigned to j^{th} machine or facility.

Now as the problem forms one to one basis or one job is to be assigned to one facility or machine.

Therefore,

$$\sum_{j=1}^n x_{ij} = 1$$

$$\sum_{i=1}^n x_{ij} = 1$$

and x_{ij} is either zero or one.

Method to solve Problem (Hungarian Technique):

Consider the objective function of minimization type. Following steps are involved in solving this Assignment problem,

1. Locate the smallest cost element in each row of the given cost table starting with the first row. Now, this smallest element is subtracted from each element of that row. So, we will be getting at least one zero in each row of this new table.

2. Having constructed the table (as by step-1) take the columns of the table. Starting from first column locate the smallest cost element in each column. Now subtract this smallest element from each element of that column. Having performed the step 1 and step 2, we will be getting at least one zero in each column in the reduced cost table.

3. Now, the assignments are made for the reduced table in following manner.

(i) Rows are examined successively, until the row with exactly single (one) zero is found. Assignment is made to this single zero by putting square $\square$ around it and in the corresponding column, all other zeros are crossed out (x) because these will not be used to make any other assignment in this column. Step is conducted for each row.

(ii) Step 3 (i) is now performed on the columns as follow:- columns are examined successively till a column with exactly one zero is found. Now, assignment is made to this single zero by putting the square around it and at the same time, all other zeros in the corresponding rows are crossed out (x) step is conducted for each column.

(iii) Step 3, (i) and 3 (ii) are repeated till all the zeros are either marked or crossed out. Now, if the number of marked zeros or the assignments made are equal to number of rows or columns, optimum solution has been achieved. There will be exactly single assignment in each row or columns without any assignment. In this case, we will go to step 4.

4. At this stage, draw the minimum number of lines (horizontal and vertical) necessary to cover all zeros in the matrix obtained in step 3, Following procedure is adopted:

- (i) Tick mark (✓) all rows that do not have any assignment.
- (ii) Now tick mark (✓) all these columns that have zero in the tick marked rows.
- (iii) Now tick mark all the rows that are not already marked and that have assignment in the marked columns.
- (iv) All the steps i.e. (4(i), 4(ii), 4(iii)) are repeated until no more rows or columns can be marked.
- (v) Now draw straight lines which pass through all the un marked rows and marked columns. It can also be noticed that in an $n \times n$ matrix, always less than 'n' lines will cover all the zeros if there is no solution among them.

5. In step 4, if the number of lines drawn are equal to n or the number of rows, then it is the optimum solution if not, then go to step 6.

6. Select the smallest element among all the uncovered elements. Now, this element is subtracted from all the uncovered elements and added to the element which lies at the intersection of two lines. This is the matrix for fresh assignments.

7. Repeat the procedure from step (3) until the number of assignments becomes equal to the number of rows or number of columns.

8. Now apply the step II to the modified matrix obtained in the above till we get the desired solution.

Unbalanced Assignment Problems

In an assignment problem, if the number of tasks is not equal to the number of persons then it is called an unbalanced assignment problem. Thus in an unbalanced assignment problem the cost matrix is not a square matrix.

To solve such a problem, it is first converted into a balanced problem by adding fictitious row and column in the matrix so that a square matrix is formed. The prices of all the elements of these additional rows or columns are taken to be zero. Then the usual assignment algorithm is applied to the balanced problem so obtained.

Problems On Assignment

1. In a computer centre after studying carefully the three expert programmes, the head of computer centre, estimates the computer time in minutes required by the experts for the application programmes as follows:

		<i>Programmes</i>		
		<i>A</i>	<i>B</i>	<i>C</i>
Programmes	1.	120	100	80
	2.	80	90	110
	3.	110	140	120

Assign the programmers to the programmes in such a way that the total computer time is minimum.

Solution:

The Hungarian method is used to obtain an optimal solution.

Step (1) & (2):

The minimum time element in row 1, 2 and 3 is 80, 80 and 110. resp. Subtract these elements from all elements in this respective row.

The reduced time matrix is shown in following table (1) Table 1:

Table 1:

	<i>A</i>	<i>B</i>	<i>C</i>
1.	40	20	0
2.	0	10	20
3.	0	30	10

In reduced Table (1) the minimum time element in columns A, B, and C is 0, 10 and 0 resp, subtract these elements from all elements in this resp. column to get the reduced time matrix as shown in Table 2.

Table 2:

	<i>A</i>	<i>B</i>	<i>C</i>
1	40	10	0
2	0	0	30
3	0	20	10

Step 3 (a):

Examine all the rows starting from first one- until a row containing only single zero element is located, Here, rows 1 and 3 have only one zero in the cells (1, C) and (3,A) resp, we assigned these zeros. All zeros in the assigned column are crossed off as shown in table 3.

Table 3 :

	<i>A</i>	<i>B</i>	<i>C</i>
1.	40	10	0
2.		0	30
3.	0	20	10

(b) We now examine each column starting from A in table 3, There is one zero in column B in the cell (2, B). Assign this cell as shown in table 4.

Table 4 :

	<i>A</i>	<i>B</i>	<i>C</i>
1.	40	10	0
2.		0	30
3.	0	20	10

(c) Since the no of Assignments (= 3) equal the no of rows (= 3), the optimal solution is obtained.

The pattern of assignment among programmers and programmes with their respective line (in minutes) is given below.

<i>Programmer</i>	<i>Programme</i>	<i>Time (in minutes)</i>
1.	C	80
2.	B	90
3.	A	100
Total = 280.		

Example 2:

A department has five employees with five jobs to be performed. The time in hours) each men will take to perform each job is given in the effectiveness matrix.

The solution is not optimal because only four assignments are made.

Step 4:

Cover the zeros with minimum numbers of lines (= 4) as explained below.

Step 5:

Develop the new revised table by selecting the smallest element among all uncovered elements by the lines in table 3 viz., 2. subtract $K = 2$ from uncovered elements including itself and add it to elements 5, 10, 8 and 0 in cells (A, 1), (A, IV), (C, 1) and (E, IV) resp. which lie at the intersection of two lines. Another's revised table so obtained is shown in table 4.

Table 4 :

	<i>I</i>	<i>II</i>	<i>III</i>	<i>IV</i>	<i>V</i>
A	7	0	8	12	11
B	0	4	13	10	1
C	10	5	0	2	0
D	0	2	0	0	3
E	3	3	4	0	6

Step 7:

Repeat step (3) to (5) to find a new solution. The new assignment is shown in Table 5.

Table 5 :

	<i>I</i>	<i>II</i>	<i>III</i>	<i>IV</i>	<i>V</i>
A	7	0	8	12	11
B	0	4	13	10	1
C	10	5		2	0
D		2	0		3
E	3	3	4	0	6

Since the no. of assignment (= 5) equals the no of rows (or columns), the solution is optimal.

The pattern of assignments among jobs and employees with their respective time (in hour) is given below:

<i>Job</i>	<i>Employee</i>	<i>time (in hour)</i>
A	II	5
B	I	3
C	V	2
D	III	9
E	IV	4
		Total = 23

(a) Mark (v) row D since it has no assignment then.

(b) Mark (v) columns I and IV since row D has zero element in these columns.

(c) Mark (v) rows B & E since column 1 and (IV) have an assignment in rows B and E resp.

(d) Since no other rows or columns can be marked draw straight lines through the unmarked rows A & C and the marked columns I and IV as shown in Table 3.

Table 3 :

	<i>I</i>	<i>II</i>	<i>III</i>	<i>IV</i>	<i>V</i>
A	5	0	8	10	11
B	0	6	15	10	3
C	8	5	0		
D		4	2		5
E	3	5	6	0	0

To convert the Maximization problem to Minimization

Sometimes the assignment problem may deal with maximization to objective function.

For example, assign persons to jobs such that the expected profit maximized.

Hungarian method work only if the objective function has to minimized. In order to convert the problem to minimization, multiply each entry by the effectiveness matrix by -1.

Travelling Salesman Problem

Suppose that a salesman wants to visit a certain number of cities allocated to him. Let the distance/cost/time of journey from i th city to j th city is denoted by C_{ij} . Now, the salesman problem is to select such a route for travelling so that the distance/cost/time is minimized.

References:

Advance Engineering Mathematics - H.K. Das, S. Chand